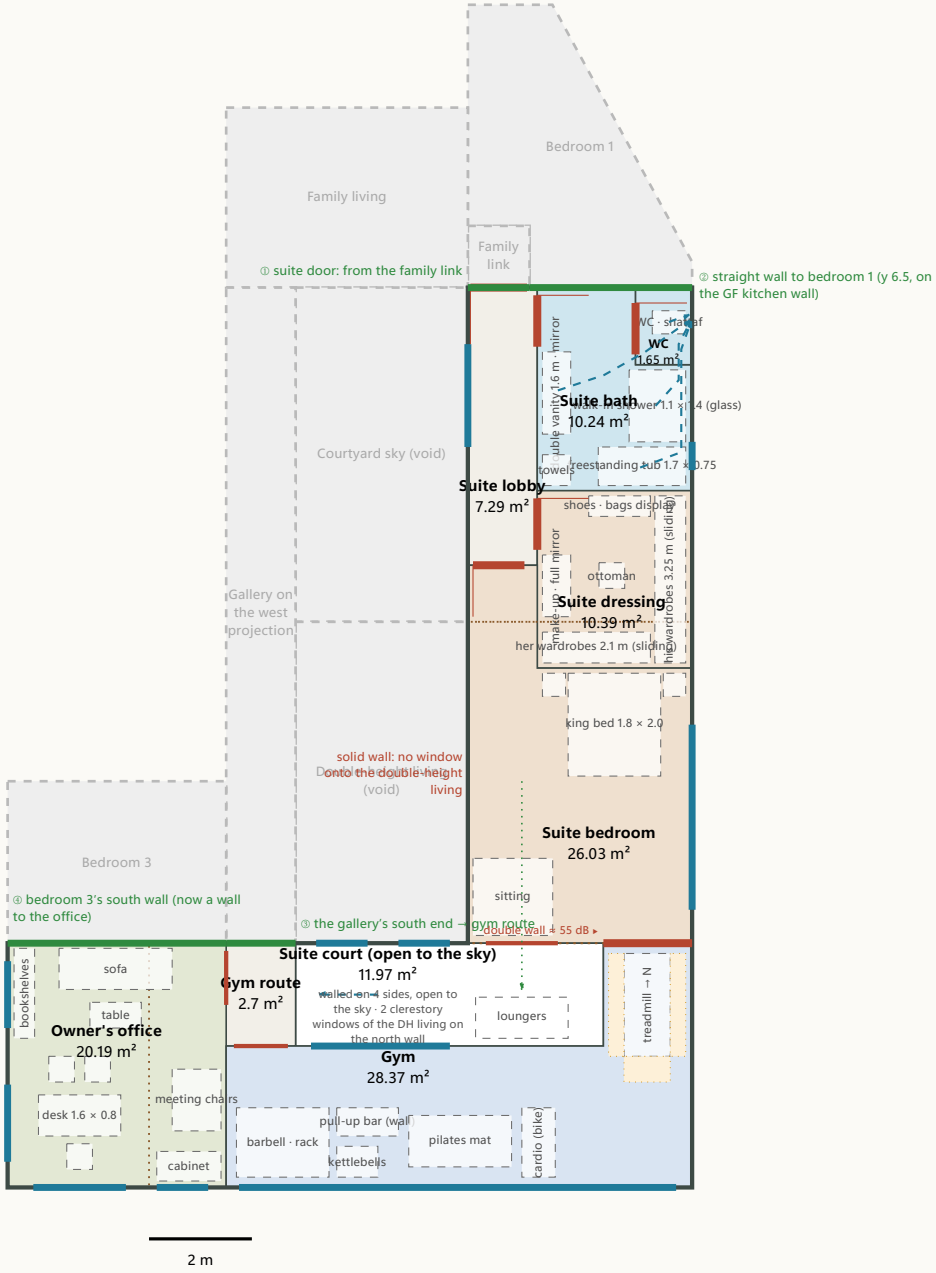
First floor · the L: owner's office → gym route → gym → court → master suite

Taif / Palm House · first floor V6.3 · CSV coordinates (x east, y south) · wall centre lines · north up

First floor V6.3 · part 2 of 2 (part 1: studies/ff_family.typ) · owner's office, L-shaped gym (treadmill arm at the east end), suite court and master suite



Rooms

Room	Axes w × d (m)	Axes m²	≈ clear m²
Suite lobby	1.35 × 5.4	7.29	5.94
Suite bath	polygon	10.24	8.79
Suite WC (closet)	1.1 × 1.5	1.65	1.09
Suite dressing	3.01 × 3.45	10.39	9.01
Suite bedroom	polygon	26.03	23.56
Suite court (open to the sky)	5.99 × 2	11.97	–
Gym route	1.35 × 2	2.7	2.03
Owner's office	4.25 × 4.75	20.19	18.16
Gym	polygon	28.37	25.26
Zone		**118.83**	

Clear ≈ axes minus half a wall on every edge (250 mm outer, 200 mm inner, nothing on open edges), corners ignored.

✓ All rooms inside the zone · no overlaps · no unassigned area · furniture inside its room, not overlapping, out of the lobby route and the gym walkway · treadmill in the arm (0.3 m each side, 0.1 m off the bedroom wall, 0.5 m free behind), gym door free · door swings clear · every room has a door reached from the lobby (or its room) · doors from the family part on the interfaces · lobby ≥ 1.0 m clear · no window onto the double-height living

Key numbers

- **Suite:** lobby 7.29 · bath 10.24 + WC closet 1.65 · dressing 10.39 · bedroom 26.03 = **55.6 m² indoors** + the suite court 11.97 m².
- **Gym** 28.37 m² · **suite court** 11.97 m² · **gym route** 2.7 m² · **owner's office** 20.19 m².
- **Walked:** suite door → bedroom door ≈ 5.4 m · bedroom door → bath door ≈ 5.5 m · bed → court ≈ 3.9 m.
- **Interfaces with the family part (green, fixed):** ① suite door at the end of the family link; ② the straight wall to bedroom 1 on y 6.5; ③ the gallery's south end, where the gym route starts; ④ bedroom 3's south wall (a wall to the office).

Plan: red = doors (thin = sliding / no leaf) with swing line · blue = windows · blue dashed = drains · green = interfaces · brown dotted = ground floor below · grey = the family part (fixed) · dashed boxes = furniture and fixtures.

Master suite, suite court, gym and office · rooms, drains, walls, questions

Suite lobby · 1.35 × 5.4 m

- The suite door (from the family link) opens into a distributor: bath, dressing and bedroom each open off it; nobody walks through the dressing. A window onto the courtyard on its west wall.
- **Bent entry:** the bedroom door is at the lobby's south end, round the corner of the dressing: the bed is not seen from the suite door.

Suite bath (luxury, compact) · 11.89 m² incl. its WC closet

- **Freestanding tub** along the south wall by an east window; **walk-in glass shower** on the east wall; **double vanity** on the west (lobby) wall facing the window light; **towel cupboard**; open floor in the middle.
- **Door** from the lobby at the north end of the west wall, opening in, swing clear; the north wall stays clear as the path from the door to the WC closet.
- **WC in its own closet** (1.1 × 1.5 m, sliding door) in the north-east corner, at the drain stack, with a shattaf.

Suite dressing (walk-in) · 3.01 × 3.45 m

- Door from the lobby at the north end of its west wall. **His wardrobes** along the east wall and **her wardrobes** along the south wall, both sliding; a shoe and bag display on the north wall; a make-up table with a full-length mirror on the west wall; an ottoman. No window: good lighting inside the wardrobes.

Suite bedroom · 26.03 m²

- **King bed** 1.8 × 2.0 on the dressing wall, facing south to the suite court; bedside tables; a sitting corner on the solid west wall.
- **West wall solid:** the double-height living is behind it, no window (checked). Windows east and the sliding door to the court (south).

Suite court · 5.99 × 2 m, walled on four sides, open to the sky

- 11.97 m², from the bedroom only (sliding door). Walled on all four sides: north the double-height living (clerestory windows) and the suite bedroom, south the gym's north wall (high obscured windows), west the gym route, east the treadmill arm's west wall (solid, no window onto the court). Fully private; it works as a light well for the clerestory windows.
- **Clerestory windows** of the double-height living (in its south wall y 19.25): their sill stays above eye level from the court floor (or obscured glass). The gym's roof edge ends in a **railing, not a parapet**, to keep the winter sun on them (light check on design_ff.pdf).
- **Open to the sky:** waterproofing on the men's majlis roof slab, the floor falling **west** to its drain (away from the treadmill arm at its east end); the clerestory windows and the bedroom door open for air in summer.

Gym · an L, 28.37 m² (long arm 9.06 × 2.75 m + treadmill arm 1.725 × 2 m)

- **Owner's choice:** the long arm along the south facade, sliding door from the gym route at its west end; the **short arm at the east end** turns north along the suite bedroom wall and holds **only the treadmill**. Windows south and high obscured windows north onto the suite court.
- **Treadmill** 0.9 × 2.0 m in the arm, facing north, **0.3 m clear each side**, 0.1 m off the bedroom wall, **0.5 m free behind it** in the long arm (the user steps off south). Arm 1.725 × 2 m on axes, 1.5 m clear.
- **Acoustics (mandatory):** a **double wall** between the arm and the suite bedroom (y 19.25, red: e.g. 200 mm dense block + an independent stud lining with mineral wool, target ≈ 55 dB airborne), sealed at the top, no sockets back to back; a **floating floor** (or at least dense rubber tiles) under the arm, stopped short of the bedroom wall; **anti-vibration pads** under the treadmill; the treadmill never touches the wall.
- **Long arm:** barbell + rack, kettlebells, pull-up bar (wall), pilates mat, one cardio machine (bike). A clear walkway runs along its north wall from the door to the arm (checked: nothing in it, the door landing free).

Owner's office · 4.25 × 4.75 m, 20.19 m² · gym route 1.35 × 2 m

- An indoor room over the GF foyer / guest passage and the men's majlis; its west wall stands on the GF west wall.
- Sofa, coffee table and bookshelves in the north half; desk with two visitor chairs, a meeting corner and a cabinet in the south half. Windows west (two) and south (two).
- **Door** (sliding, lockable) from the gym route, just past the end of the gallery. The gym route serves the office and the gym.

Drains (blue dashed) · no east duct

- **Suite bath:** all fixtures drain east in a **sunken bath floor** to one **stack on the outside face of the east wall**, down to the east-yard manhole where the kitchen already drains. Nothing hangs in the GF kitchen ceiling; the hood on the GF east wall stays clear of the stack (south of y ≈ 7.5).
- No drain over the master bedroom. The suite court drains to its own floor drain.

Walls (engineer)

- **On GF lines:** y 6.5 (bedroom 1 / suite: the GF kitchen's north wall), the outer walls, x 8.95 (courtyard / double-height living). The suite bedroom spans the GF kitchen / living wall (y 13.0) with no wall above it.
- **Off the GF lines:** x 10.3, y 10.45, x 12.211 and y 8.0, y 11.9 and y 13.9, all over the GF kitchen; on the men's majlis roof: y 21.25 (gym / suite court), x 4.25 (office / gym route and gym), x 5.6 (gym route / suite court), **x 11.586** (the treadmill arm's west wall, no wall below, with the treadmill's moving load ≈ 1.5–2 kN). The arm's north wall y 19.25 is on the GF living / men's majlis wall line.
- **Loads and vibration** of the gym (treadmill) and the office over the men's majlis: engineer.

Questions for the owner

1. **Bedroom:** king bed facing the court, or another arrangement (TV wall, sitting area)?
2. **Suite court:** 11.97 m², long and narrow: enough? Clerestory privacy (sill height or obscured glass) to settle.